

REJOINDER:

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We would like to thank our respondents for their time and insightful comments on this paper. Their comments have certainly brought out new aspects in the interpretation of information measures in general and the Shannon-Lindley measure in particular.

The first point we would like to touch on is the concept of negative information, questioned by Stewart and Geertsema. As stated by Geertsema, a negative value of I_x indicates that the probability is more dispersed in the posterior than in the prior or, according to Stewart, there is an increase in confusion in the mind of the observer. This is a feasible interpretation within the Bayesian paradigm (if unsatisfactory to the observer). Nor is the observer necessarily worse off. There may be a reduction in the information concerning the value of the parameter, but this is an indication that the observer was too sure of himself *a priori*, and leads to a more realistic posterior opinion. Stewart is certainly correct in not wanting to blame the data for negative information. Granting the correctness of the sampling technique, the prior should shoulder the blame and the data is merely making a correction to an over-confident prior.

In the Bernoulli example with a two point prior, Geertsema shows that the observation may contradict the prior and give either negative or positive information. This is because the information measure only compares the dispersions of the prior and posterior distributions. In the first example $p_1 = 0,3$, while in the second $p_1 = 0,4$, so that the prior is "weaker" in the second and therefore it is more likely that the observation will contribute positive information. When $p_1 = 0,5$, I_x will always be positive, irrespective of the value of the observation. In general there will always be an interval for p_1 where the observation contradicts the prior while yielding positive information.

The comparison made by Stewart between I_x and J_x (the Kullback-Leibler measure) is very interesting and it is clear from figure C1 that these two

quantities measure different concepts of information. We agree with his view of $I_{\bar{x}}$, namely the reduction in confusion regarding the value of the parameter, but would like to put a different perspective on the meaning of $J_{\bar{x}}$. The latter is basically a measure of the divergence between two distributions; it is a measure of the amount of *change* between prior and posterior opinions, particularly change in location. For example, if the observer has a $N(0, 1)$ prior and, after receiving some information, changes that to a posterior that is $N(4, 1)$, he has undergone a massive change of opinion regarding the likely value of the parameter. This will be reflected by a large value of $J_{\bar{x}}$. However, he is no more confident about his new opinion than he was about his original one, and this is illustrated by the fact that $I_{\bar{x}} = 0$. These two measures are complementary, but in the sense of Shannon and Lindley, information as the reduction of entropy, $I_{\bar{x}}$ is the appropriate one. It is interesting to note that for the binomial model (Example 3) with uniform prior $I_{\bar{x}}$ and $J_{\bar{x}}$ are identical. For the other examples the profile of $J_{\bar{x}}$ is similar to figure C1.

An important point raised by Fatti and Geertsema is the sensitivity of the measure to prior specifications. In the life testing example, as in the example of figure 6, bounded parameter space and unbounded sample space cause the rather unnatural increase in $I_{\bar{x}}$ beyond the range of the parameter since all the posterior density is then concentrated against the boundary of the parameter space and this should indicate that something is wrong, probably in the prior specifications. Perhaps the answer is to look for a class of priors with respect to which the measure is reasonably robust. The normal prior is however the other extreme and is too insensitive, the information measure being independent of the data.

Crowther gives an interesting measure by changing the operator of the Fisher information around. According to the interpretation of the Fisher information statistic, this measure should then describe the sensitivity of the posterior to changes in the data.